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Improved DC Properties of Polypropylene HVDC Cable Insulation by Grafting and Blending Combined Modification

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ABSTRACT

In order to meet the needs of mechanical properties of cable insulation, polypropylene (PP) needs to be blended with low modulus materials to achieve toughening and softening. However, the compatibility between two or more resins can significantly affect the electrical properties of the composites. To solve this problem, the styrene–ethylene/butylene–styrene block copolymer (SEBS) with high heat resistance is selected as the blending elastomer, and grafted polypropylene (GPP) is prepared to be as basic resin. The morphological structure, thermal properties, mechanical properties and high temperature DC electrical properties of PP, GPP, PP/SEBS and GPP/SEBS are studied. The results demonstrate that compared with PP/SEBS, GPP/SEBS has significantly refined ‘island phase’ structure, further reduced storage modulus and slightly increased tensile strength and elongation at break and higher viscosity at low shear frequency. Electrical properties test results show that the grafted composite exhibits obvious suppression effect on space charge and conductance current, and enhances the DC breakdown strength. The introduction of the grafted material not only effectively improves the compatibility of the composites but also significantly improves the electrical properties of the material. This result can provide a reference for the development of eco-friendly high voltage cable materials.

1 | Introduction

With the rapid development of new power systems and high voltage direct current (HVDC) transmission, HVDC cables face the requirements of higher operating voltage grades and higher operating temperature [1–3]. As a representative of thermoplastic materials with excellent electrical properties and high crystallisation and melting temperature, polypropylene (PP) has gradually become a new generation eco-friendly insulation material to replace cross-linked polyethylene (XLPE) [4–7]. However, because of the regular arrangement of the micro-molecular

structure of PP, when it is subjected to mechanical stress, the crystallisation zone will develop cracks and gradually spread and transfer, which is easy to brittle fracture and has high hardness. Toughening modification by blending or copolymerising with low modulus elastomer is an effective strategy, and materials with high elastic properties are dispersed in the PP matrix as stress concentration points, which can improve the mechanical properties of the composites. It carries most of the applied stress and causes crazes and shear bands, absorbs the energy generated by the external impact, improves the impact strain capacity and effectively reduces the hardness of the material [8–10].

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Ethylene–octene copolymer (POE) and propylene–octene copolymer (POP) are commonly used as toughening elastomers for blending modification [11], but their melting temperatures are much lower than PP. At the operating temperature of the cable, the melt with extremely low electrical strength will be generated in the blend material, which may cause potential safety hazards when used as HVDC cable insulation [12]. Impact copolymerised polypropylene (IPC) with improved toughness was prepared by introducing amorphous ethylene–propylene elastomer phase into PP through in situ polymerisation. Its remarkable advantage is that the elastomer phase introduced by copolymerisation has strong binding force and uniform dispersion with PP phase [13–16]. But similar to POE and POP, the elastomeric phase is a melt at high temperatures, so it undergoes the same electrical degradation at high temperatures.

The blending material with styrene ethylene butylene styrene (SEBS) as toughening filler can effectively reduce the modulus of the material on the premise of ensuring the relatively high heat resistance temperature, and the physical crosslinking network formed by the polystyrene structural unit can improve phase structure stability of the composite. In addition, the polystyrene segment on the SEBS can introduce charge traps to improve the DC electrical properties, so the PP/SEBS composite is expected to achieve synergistic improvement of physical properties and DC electrical properties [17, 18]. However, PP/SEBS still faces two problems. The first is the homogeneity of the material. Because of the existence of polystyrene structural units, SEBS has a certain polarity, and a simple blending process cannot achieve homogeneous blending. Besides, due to the large melt viscosity of SEBS, when the aggregated SEBS drops exist in PP/SEBS, the viscosity of the composite will be too high to be extruded.

In order to solve the problem of PP/SEBS homogeneous blending, ultraviolet (UV) irradiation grafting technology with high efficiency was used to prepare grafted PP (GPP) to improve the dispersion of SEBS. PP, GPP, PP/SEBS and GPP/SEBS composites were prepared, respectively. The microstructure, physical properties, viscosity and high-temperature DC electrical properties of the materials were tested and the influence mechanism of grafting modification on the properties of the composites was comprehensively evaluated.

2 | Materials and Experiment

2.1 | Materials

The base resin PP (grade K8003) was purchased from Sinopec Yanshan Petrochemical Co. The melting point is about 167°C, and the melt index is 2.2 g/10 min. Photo initiator benzophenone (BP), grafted monomer maleic anhydride (MAH) and functional second grafted monomer 4-tert-butyl styrene (St*) were purchased from Aladdin Biochemical Technology Co.

The high-performance grafting masterbatch was prepared by UV irradiation initiation method [19]. The weight ratio of PP, BP, MAH and St* was 100:0.2:3:1, and the 300 µm flakes were prepared after blending at 180°C. The molten material was then

irradiated for 2 s under a 365 nm light-emitting diode UV light array to initiate the grafting reaction and the GPP masterbatch was obtained.

PP, GPP, PP/SEBS and GPP/SEBS composite blending compounds are shown in Table 1. The composites were prepared by melt blending at 180°C, then pressed and moulded at 180°C, 15 MPa, water-cooled and then vacuum short-circuited in a drying oven at 80°C for 48 h for subsequent performance characterisation. It is worth noting that in order to avoid the difference of electrical properties caused by different grafting contents and ensure the comparability of materials, the grafting ratio of GPP and GPP/SEBS is controlled at the same content in the ratio.

2.2 | Molecular Structure

In order to determine the content of the grafted monomer in the material, a Fourier transform infrared (FTIR) spectrometer produced by Gasco (Shanghai) Co., Ltd. was employed. The test spectral wavenumber range is 2000 to 1500 cm⁻¹, the scanning accuracy is 2 cm⁻¹, and the scanning times are 30.

2.3 | Phase Distribution and Compatibility Simulation

Hitachi SU8020 cold field emission high-resolution scanning electron microscope (SEM) was used to observe the dispersion of SEBS in PP. Before observation, the sample was brittily fractured at 80 K in liquid nitrogen, and then the section of the 1-mm-thick sample was etched by n-heptane at 50°C for 20 min. The brittle section was sprayed with gold and observed.

The compatibility parameters of PP/SEBS and GPP/SEBS systems were calculated by the Forcite software package on Materials Studio. PP and SEBS with a polymerisation degree of 20 were established; the GPP molecular model was 1 St*, and MAH molecules were grafted onto PP molecular chain. The geometry optimisation module was used to optimise the structure of the model, and then the mixing component in the blends module was used to calculate and analyse the composite. The computational basis for simulation was COMPASS II.

2.4 | Thermal and Mechanical Properties

The crystallinity and crystallisation-melting temperature of the composites were measured by Mettler Toledo DSC822e. A

TABLE 1 | Compounds of PP and modified PP.

Materials	Content (wt%)		
	PP	GPP masterbatch	SEBS
PP	100	—	—
GPP	30	70	—
PP/SEBS	70	—	30
GPP/SEBS	—	70	30

sample about 5 mg was first heated from 25°C to 200°C and held at constant temperature for 5 min to eliminate the effect of the thermal history of the material. After that, the sample was cooled to 25°C and then heated to 200°C to record the melting curve. The test was carried out in a high-purity nitrogen atmosphere with a flow rate of 50 mL/min. The cooling and heating rate is 10°C/min.

The dynamic viscosity of the samples was measured by MCR302 rotational rheometer made in Anton Paar, Austria. A 25 mm parallel plate clamp was used, the shear frequency range was 0.05–100 rad/s, the test temperature was 230°C, and the sample was preheated for 8 min before the test, the number of sampling points was 67.

The dynamic thermo-mechanical properties of the material were measured by Q800 from TA Instruments. The sample was cut into the standard size of 9 mm × 4 mm × 1 mm, and the experimental mode was tensile mode. The parameters were set as follows: a target amplitude of 15 µm, a vibration frequency of 1 Hz, a static force of 0.375 N and a dynamic force of 0.3 N.

The stress–strain curve was carried out on the CMT6000 multi-function electronic tensile machine produced by Meishan Industrial Systems Co. The 1 mm sample was cut according to the 5A standard type, and the tensile speed was 50 mm/min. Each group of samples is tested 5 times, and the data with high repeatability are taken as the final test result.

2.5 | DC Properties

Based on pulse electroacoustic (PEA) method, space charge characteristics of materials were tested. The thickness of the sample was (400 ± 10) µm. The polarisation electric field was set as 40 kV/mm and the temperature was 80°C (maximum temperature of the PEA test platform). The space charge distribution during polarisation for 7200 s was recorded. Then the polarisation electric field was removed, and the space charge distribution during depolarisation for 600 s was recorded.

The conductance current of the material was tested at 80°C using a sample with a thickness of 200 µm and a three-electrode test device, with the conductance current measured from 10 to 45 kV/mm by a picoammeter. The quasi-steady state current values for each field polarisation lasting for 60 min were recorded and the corresponding curves were plotted.

The breakdown voltage at 90°C was measured. The thickness of the test sample was about 100 µm, and the high voltage and grounding electrodes were cylindrical electrodes with diameters of 25 and 70 mm, respectively. During the test, the sample and electrode were immersed in dimethyl silicone oil. The voltage rise rate was 500 V/s. Finally, the two-parameter Weibull distribution was used to analyse the experimental data, and the corresponding breakdown strength and shape parameter were obtained.

2.6 | Trap Distribution

Thermally stimulated depolarisation current (TSDC) was used to characterise the trap distribution in the material. The thickness of the sample was 100 mm, the polarisation temperature was 80°C, the polarisation electric field was 40 kV/mm, and the polarisation time was 30 min. After polarisation, liquid nitrogen was used to cool the sample to about −50°C, and the sample was short-circuited for 3 min to release the discharge current caused by instantaneous polarisation. Then the sample was heated to 165°C at a rate of 3°C/min and the depolarisation current was collected.

Based on the first principles, the density of states (DOS) of four groups of molecular models were calculated by using the Dmol3 package of Materials Studio. The degree of polymerisation of the molecular model is 20, and the calculation is carried out under the B3LYP basis set [20].

3 | Results

3.1 | Molecular Structure

FTIR results are shown in Figure 1. After grafting modification, a new characteristic absorption peak appears at the wavenumber of 1783 cm^{−1} in GPP, which corresponds to the stretching vibration peak of the C=O group on the five-membered cyclic anhydride in MAH molecule [19, 21]. It indicates that MAH was successfully grafted onto PP macromolecular chains by UV irradiation. The PP/SEBS material exhibits new characteristic absorption peaks at 1601 and 1583 cm^{−1}, which are caused by the flexural vibration of benzene functional groups on the styrene segment in SEBS.

3.2 | Phase Distribution and Compatibility Simulation

3.2.1 | SEM

The SEM images of each group of samples are shown in Figure 2. The holes in Figure 2a,b represent the ‘Rubber phase’

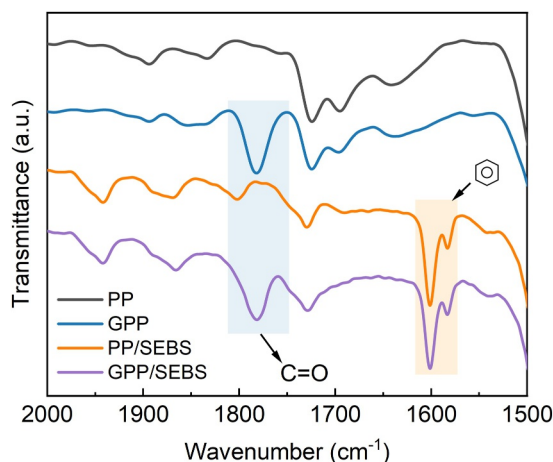


FIGURE 1 | FTIR spectra of PP and modified PP.

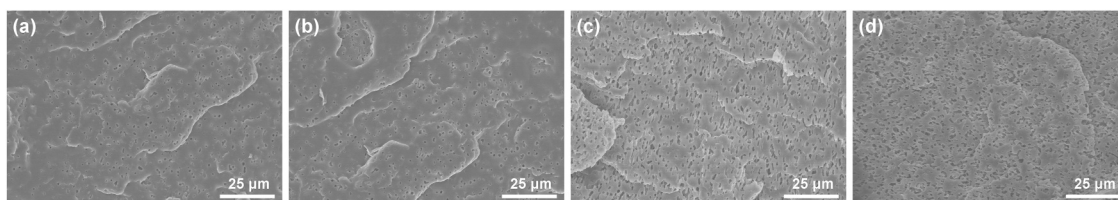


FIGURE 2 | SEM images of PP and modified PP. (a) PP, (b) GPP, (c) PP/SEBS, (d) GPP/SEBS.

in PP. It can be seen that there is no obvious difference between PP and GPP, and the grafting modification does not change the morphology of the material significantly. The voids in the Figure 2c,d are formed by etching elastomer SEBS dispersed in the PP. The blend system showed a 'sea island' phase structure with SEBS as the 'island' phase and PP as the 'sea' phase. The PP/SEBS blends show an impaired dispersibility, and the SEBS regions are interconnected and large in size, with a maximum span of about 12.2 μm . In contrast, the size of the GPP/SEBS dispersed phase is significantly reduced, and the maximum span is reduced to 3.2 μm , with a better dispersion characteristic. It can be seen that grafting polar groups onto the PP molecular chain can achieve an ideal dispersion effect, significantly improve the compatibility between PP and SEBS, and effectively improve the homogeneous degree of the blend system.

3.2.2 | Compatibility Simulation

The compatibility of PP/SEBS and GPP/SEBS was calculated according to Flory–Huggins theory. The results are shown in Figure 3. χ is the interaction parameter, which reflects the change of interaction energy in the blending process. The smaller the change, the better the compatibility of the two materials. The free energy is the energy difference between the interaction free energy of pure and mixed molecules. The smaller the energy difference is, the easier it is to exist in the material. As can be seen from Figure 3, the interaction parameters and free energy of GPP/SEBS are lower than those of PP/SEBS at various temperatures, indicating that GPP and SEBS have better compatibility after grafting modification. The calculation results are in agreement with the results of morphology characterisation in Figure 2.

3.3 | Thermal and Mechanical Properties

3.3.1 | Melting and Crystallinity Characteristics

The melting temperature and crystallinity results of the materials are shown in Figure 4. X_c is the crystallinity, which indicates the degree of structural order inside the material. Comparing PP with GPP, it can be seen that GPP shows a new melting peak and the crystallinity decreases slightly, which is related to the inevitable degradation of macromolecular chains during the grafting reaction. After blending modification, the melting temperature of PP/SEBS decreases slightly compared with that of PP, whereas the crystallinity decreases relatively significantly. The results show that the composite has a similar heat resistance to PP and the flexibility of the material is improved obviously. The crystallinity of GPP/SEBS composites decreased further, which was attributed to the lower crystallinity of GPP itself on the one hand and the

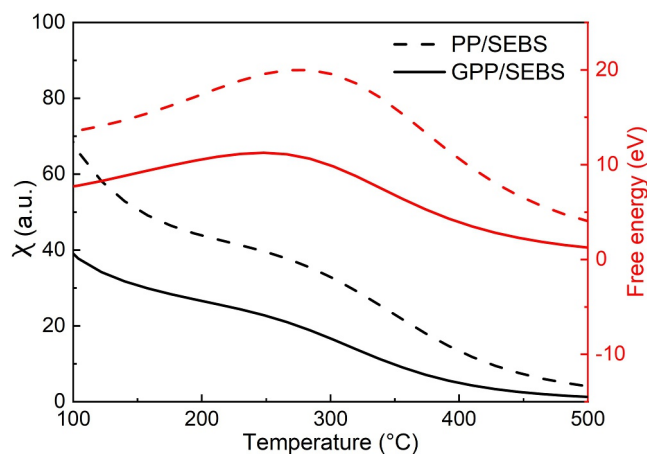


FIGURE 3 | The curve of compatibility parameters of composites with temperature changing.

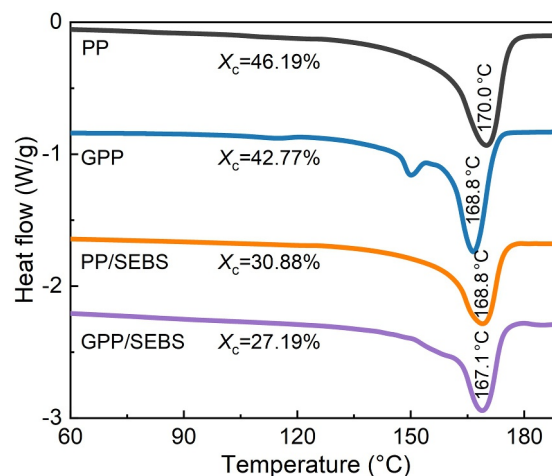


FIGURE 4 | Differential scanning calorimetry (DSC) melting curves of PP and modified PP.

more uniform dispersion of SEBS on the other hand. In addition, blending with SEBS weakened the melting peak of GPP caused by the degradation of the material at a lower temperature and improved the thermal stability of the material.

3.3.2 | Viscosity Characteristics

The complex viscosity versus shear frequency at 230°C is shown in Figure 5. Compared with PP, GPP after grafting modification has a higher shear viscosity. This difference is obvious at low frequencies and gradually weakens at high frequencies. The reason is that the grafting process is caused by the introduction

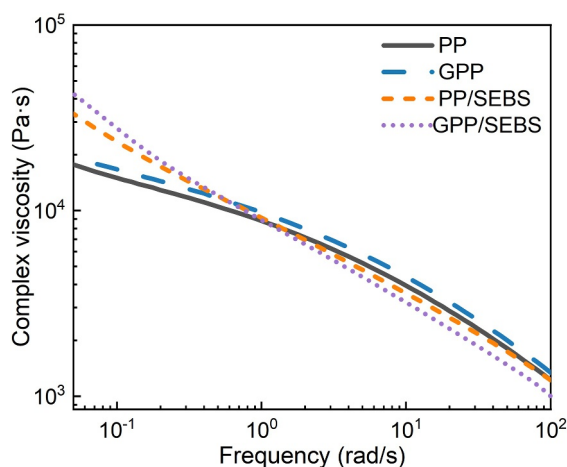


FIGURE 5 | Complex viscosity curve of PP and modified PP.

of side groups into PP molecular chains to form short branched chains. After compounding with SEBS, PP/SEBS shows a larger viscosity at low shear frequencies, and the viscosity of GPP/SEBS increases further at low shear frequencies, showing a more significant shear thinning behaviour.

The end group of the elastomer SEBS is the hard segment of polystyrene, which is easy to form physical crosslinking points, and the composite shear viscosity of SEBS's composites will increase at a lower shear frequency. Moreover, the relatively small molecular weight makes it prone to shear thinning behaviour, resulting in smaller shear viscosity of PP/SEBS in the high frequency region. After grafting modification, the interphase dispersion of GPP/SEBS is more uniform, and the rubber phase is small in size and large in number, resulting in more physical crosslinking points generated, which further increase its shear viscosity at low shear frequencies. In addition, the better dispersity reduces the size of SEBS aggregates and reduces the shear viscosity at high frequencies. GPP/SEBS shows lower shear viscosity and higher shear viscosity at low shear frequencies, which is beneficial to cable insulation moulding extrusion and eccentricity control.

3.3.3 | Stress–Strain Curve

The stress–strain results are shown in Figure 6. It can be seen that unmodified PP has a tensile yield stress of up to 18.25 MPa. After grafting modification, the yield stress of GPP is slightly lower than that of PP, and the elongation at break is almost unchanged. In contrast, after blending modification, the tensile yield stress of PP/SEBS composites is significantly reduced, and the tensile strength is significantly improved. After grafting and blending, the tensile yield stress of GPP/SEBS materials is similar, and the tensile strength and elongation at break are increased to 42.7 MPa and 547.5%, respectively. It indicates that the grafting of PP plays a positive role in improving the dispersion of SEBS. On the one hand, the compatibility between GPP and SEBS is better, and the phase distribution of the composite is improved. On the other hand, the intermolecular force between GPP molecules and SEBS is enhanced, and the tensile strength and elongation at break of the GPP/SEBS composite are further increased.

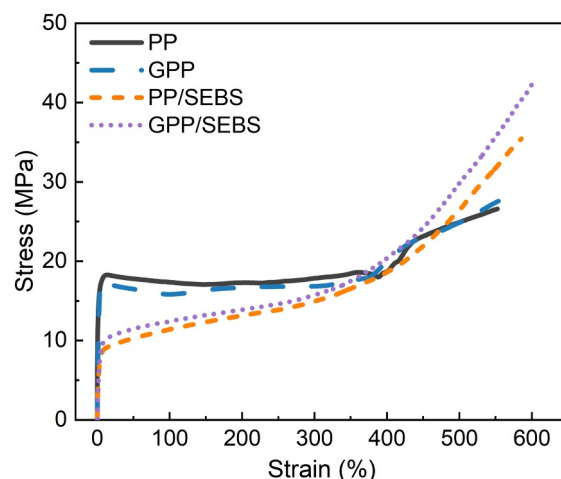


FIGURE 6 | Stress–strain curves of PP and modified PP.

3.3.4 | Dynamic Mechanical Analysis (DMA)

The DMA results are shown in Figure 7. Compared with PP, the storage modulus of GPP decreases slightly near 0°C, and the peak value of loss factor increases. It can be attributed to the introduction of side groups on the PP macromolecular chain by grafting modification. On the one hand, the relaxation of side groups will cause the rise of loss peaks; on the other hand, grafting will reduce the regularity of PP, so that its crystallinity will decrease and its modulus will decrease, which is consistent with the DSC results in Figure 4. After blending with SEBS, the storage modulus of the PP/SEBS and GPP/SEBS decreases significantly compared with PP. At ambient temperature, the storage modulus of PP is 1360 MPa, while that of PP/SEBS decreases to 770 MPa and that of GPP/SEBS decreases to about 590 MPa. The introduction of elastomeric SEBS can significantly reduce the rigidity of the material and improve the elasticity of the composite. The toughening effect is more remarkable. As the loss factor ($\tan \delta$) curves shown in Figure 7b, PP/SEBS composite retains the β relaxation peak at about 10°C and the α relaxation peak at about 80°C inherent in PP, whereas a distinct loss peak appears at about −50°C, which caused by the glass transition process of the SEBS phase. Compared with PP, the loss peak position of the two groups of composites did not change significantly near −50°C, whereas the peak value increased. The loss peak near 80°C corresponds to the α relaxation process of the material, which is mainly caused by the movement of molecular segments on the surface of lamellae, that is, the movement of short segments of the folded region between lamellae and the tail chain part. Compared with PP and PP/SEBS, the loss peak of GPP/SEBS composites increases at higher temperature, which is related to the increased interphase bonding of grafted MAH.

3.4 | DC Properties

3.4.1 | Space Charge Distribution

Space charge distribution of four groups of materials at 80°C is shown in Figure 8. At elevated temperatures, the kinetic energy of the charge increases, the relaxation of the molecular chain is intensified, and a considerable part of the bound charge trapped

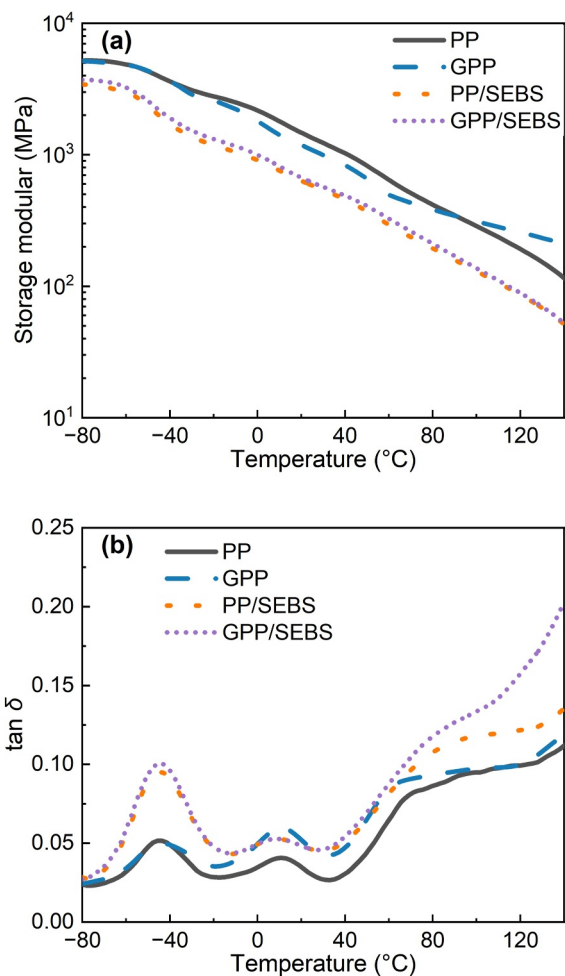


FIGURE 7 | DMA results of PP and modified PP. (a) Storage modular curves, (b) $\tan \delta$ curves.

in the shallow trap is released. Therefore, the space charge injection and accumulation characteristics of the material can be better demonstrated at high temperature. As can be seen from Figure 8a, a large amount of homocharge injection occurs in the PP at the initial stage, and as the polarisation process continues, the positive charge packet gradually migrates and accumulates near the cathode. After grafting modification, GPP in Figure 8b only shows charge injection behaviour, and no obvious charge packet migration can be observed. In Figure 8f, the charge near the electrode decays rapidly during the short-circuit process in the GPP, while the charge decays slowly in the middle of sample. As shown in Figure 8c,d, the accumulation of heteropolar charge in PP/SEBS sample is more significant. The GPP/SEBS after grafting modification shows excellent space charge suppression effect at high temperature. There is no charge packet migration inside the material, and only a small amount of positive charge injection occurs. The amount of charge accumulation in GPP/SEBS is the least. As shown in Figure 8e–h, space charge in PP/SEBS decays more rapidly than PP with the addition of SEBS, whereas the decay rate of space charge in GPP/SEBS is between GPP and PP/SEBS. During the short-circuit depolarisation process, the charge release rate is significantly related to the trap depth of the trapped charge. Therefore, the results show that SEBS can introduce charge traps, but the trap depth is not as deep as the graft modification. Comprehensively comparing the four groups of samples, it can be seen that the GPP/SEBS of graft modification and blend SEBS shows a synergistic improving effect on the space charge suppression of PP insulation.

3.4.2 | Conductance Current Characteristics

The conductance currents versus electric field curves of the four materials at 80°C are shown in Figure 9. It can be seen that compared with PP, PP/SEBS's conductance current exhibits suppressing effect at low electric field region, and the suppressing

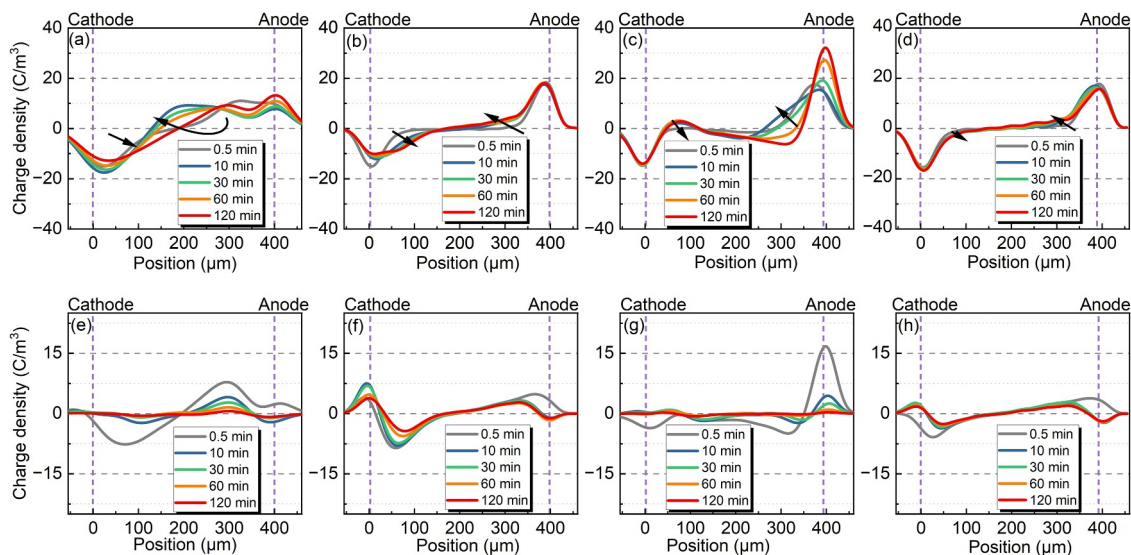


FIGURE 8 | Space charge distribution of PP and modified PP at 80°C. (a), (b), (c) and (d) are the space charge distribution during polarisation process under 40 kV/mm of PP, GPP, PP/SEBS and GPP/SEBS, respectively. (e), (f), (g) and (h) are the space charge distribution during depolarisation process of PP, GPP, PP/SEBS and GPP/SEBS, respectively.

effect gradually decreases with the increase of electric field. In contrast, the effect of graft modification on the suppressing of conductance current is more obvious. At 20 kV/mm, the GPP/SEBS conductance decreases 600 times and the GPP/SEBS conductance decreases 30 times. The conductance mechanism of polymer is complex, which contains both electronic conductance and ionic conductance. Compared with PP, the decrease of PP/SEBS current may be due to the trapping of impurity ions by the traps formed by SEBS. Compared with GPP, the increase of GPP/SEBS's conductance current may come from the intrinsic conductance current of the elastomer much higher than that of PP. Even so, GPP/SEBS produces a sufficiently strong suppression of the conductance current.

3.4.3 | Breakdown Strength

Figure 10 shows the Weibull statistics for the DC breakdown strength at 80°C for four groups of materials. It can be seen from

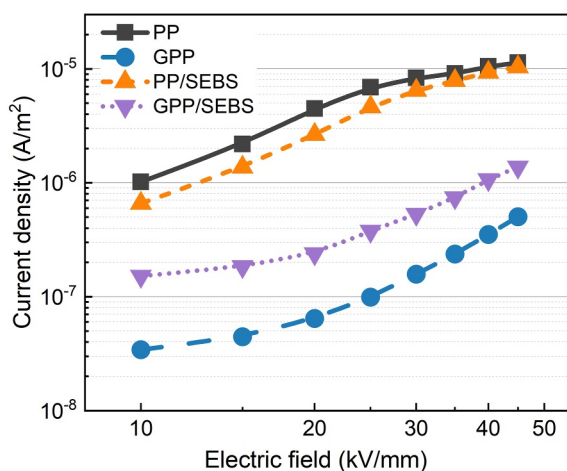


FIGURE 9 | Conductance current curves of PP and modified PP at 80°C.

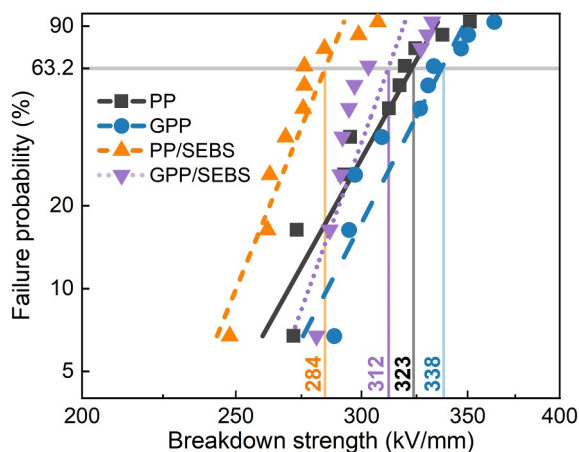


FIGURE 10 | Weibull distribution of DC breakdown strength of PP and modified PP at 80°C.

the test results that compared with PP, the breakdown field strength of GPP after grafting modification increased by 4.6%. The characteristic breakdown strength of PP/SEBS is obviously lower than that of pure PP. Compared with PP/SEBS, the GPP/SEBS composite sample shows a significant improvement in breakdown strength by 9.9%, which is significantly higher than that of PP. On the one hand, the improvement of breakdown strength is due to the better dispersion effect of SEBS in GPP, and on the other hand, it is related to the more reasonable trap distribution characteristics after GPP and SEBS are compounded. In addition, after grafting modification, the interface energy between PP matrix resin and SEBS decreased, and the intermolecular force was enhanced, which also played a positive role in increasing the breakdown strength of composites. The DC breakdown strength is significantly affected by the conductance current, and the results in Figure 10 agree well with the conductance current test results in Figure 9.

3.5 | Trap Distribution

3.5.1 | Thermal Stimulation Depolarisation Current

The TSDC test results are shown in Figure 11. It can be seen that PP sample has two obvious charge release peaks at 80°C and 148°C. The charge release peak at 80°C corresponds to the α relaxation process of the PP, and the intrinsic trap charge release caused by the structural defects in the crystal region is consistent with the DMA result in Figure 7. The charge peak at 148°C is the crystallisation pre-melting peak, which is mainly due to the charge release process caused by the rotation and relaxation of the molecular chain in the crystalline phase at elevated temperature. Compared with PP, the peak position of the intrinsic trapped charge release peak in PP/SEBS does not change significantly after introduced elastomer SEBS, but the peak value is reduced. In addition, a new charge release peak appears at 44°C in PP/SEBS, and the temperature of the charge peak is lower than the intrinsic charge release peak of PP. Therefore, the charge release peak

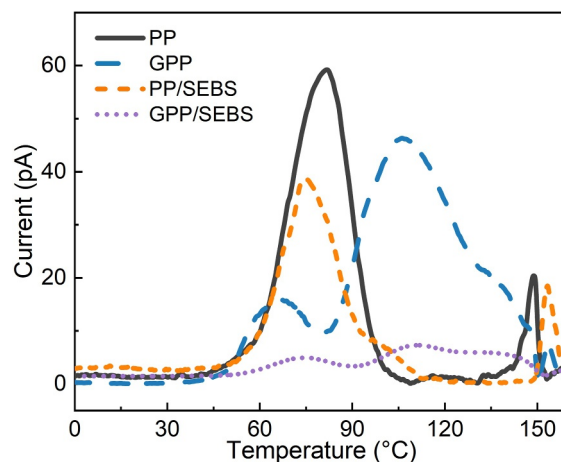


FIGURE 11 | TSDC curves of PP and modified PP at 80°C.

introduced by SEBS can be considered as shallow trap, which is consistent with the PP/SEBS space charge characteristic in Figure 8g. After grafted, GPP/SEBS shows a charge release peak at higher temperature, which is caused by the deep traps introduced by the grafted functional groups. In addition, the lower charge release peak exhibited by GPP/SEBS implies that it accumulates less space charge during the pressurisation phase, which corresponds to the space charge test results in Figure 8d.

3.5.2 | Density of State

The DOS results for the four materials are shown in Figure 12. There are no localised states in the unmodified PP, and the band gap is 7.37 eV. After graft modification, new local state energy levels appear in the GPP band gap, where the hole traps near the valence band are up to 2.82 eV, and the electron traps near the conduction band are up to 3.05 eV. After blending with SEBS, obvious electron traps appeared in PP/SEBS with a depth of 1.49 eV, which was less than that of GPP. In GPP/SEBS, both the deep charge traps in GPP and the shallower charge traps in SEBS are retained. The simulation results are consistent with the DC performance test results.

4 | Discussions

Performance comparison of PP and modified PP is shown in Table 2. In the PP/SEBS, the incorporation of elastomer SEBS as a second phase increases the nucleation point and molecular interface of PP, destroys the lamellae growth, inhibits the free crystallisation process of PP molecular chains, makes the crystallisation imperfect and improves the mechanical properties of PP/SEBS composites, reducing the crystallinity of PP materials. Although SEBS can affect the crystallisation process of PP, it

does not affect the formation mechanism of PP crystallisation. SEBS only exists in the amorphous phase of PP as a filler phase, and the morphological structure phase transition of PP is still an independent behaviour. Thus, PP/SEBS maintains a higher melting temperature similar to that of PP, as shown in Figure 4. When the material is subjected to external force, compared with the pure PP material, the elastomer particles in the composite system become the main stress concentration point, which effectively controls the further development of the craze and avoids its deterioration into destructive cracks. At the same time, when the shear band interacts with the crazing or the crazing meets with each other, the steering and branching process can effectively alleviate the damage of the material, and the toughness of the material is significantly improved and elongation at break is increased, as shown in Figure 6.

Although the blending of SEBS significantly improves the mechanical and physical properties of the material, there are still problems in the direct application of PP/SEBS composites to HVDC cable insulation. One is that the uniformity of mechanical properties cannot be guaranteed because of the large dispersion scale of SEBS and the weak consistency of composites under 'phase separation'. Another is that, because of the existence of SEBS polystyrene hard segment, physical cross-linking points can be formed in the amorphous region of the material, which increases the shear viscosity and is not conducive to processing. The DC electrical properties of its three semicrystalline polymers are closely related to the crystal structure, and after the addition of elastomer SEBS to PP, the crystalline structure of the material as a whole becomes loose, resulting in an increase in the conductivity current, which can be used to improve the electrical conductivity of PP. Moreover, the non-uniform 'sea island' phase structure interface can also provide channels for carrier transport, so the PP/SEBS material has the breakdown strength drop shown in Figure 10. The root of the problem is how to improve SEBS dispersion.

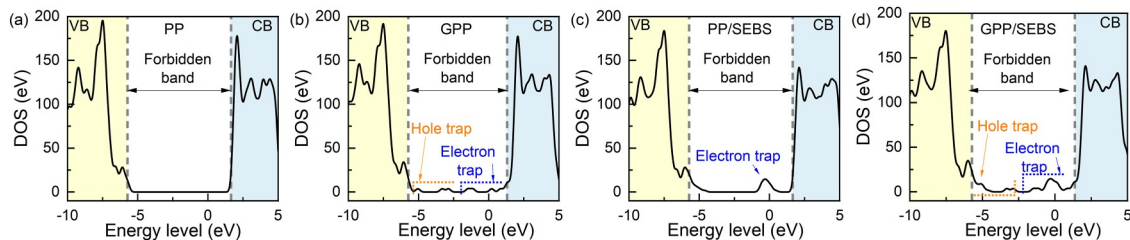


FIGURE 12 | DOS calculation results of (a) PP, (b) GPP, (c) PP/SEBS and (d) GPP/SEBS.

TABLE 2 | Performance comparison of PP and modified PP.

Materials	Melting temperature (°C)	Complex viscosity at 0.05 rad/s and 200°C (Pa·s)	Tensile strength under 50 mm/min (MPa)	Peak space charge density at 80°C and 40 kV/mm (C/m ³)	Current density at 90°C and 40 kV/mm (A/m ²)	Breakdown strength at 90°C (kV/mm)
PP	170.0	1.7×10^4	26.6	11.5	1.1×10^{-5}	323
GPP	168.8	1.9×10^4	28.5	5.2	5.0×10^{-7}	338
PP/SEBS	168.8	3.3×10^4	35.4	7.1	1.0×10^{-5}	284
GPP/SEBS	167.1	4.2×10^4	42.7	3.8	1.3×10^{-6}	312

The dispersion of SEBS in GPP/SEBS was significantly improved by UV grafting modification, and the mechanical and physical properties were slightly improved after the introduction of polar group side groups on PP and the intermolecular interaction between GPP and SEBS was enhanced. Therefore, the GPP/SEBS curve in Figure 6 is similar to the PP/SEBS curve at the early stage of stretching, but shows the effect of extending the stress hardening segment at the later stage. In terms of viscoelastic properties, the improved dispersion implies the existence of a larger number of polystyrene hard segment structures with smaller aggregation sizes and physical crosslinking points in GPP/SEBS. At lower shear rates, the existence of these abundant polystyrene hard segment structures and physical crosslinking points more effectively restricts the molecular chain motion, whereas when the shear rate is high, the hard segment structure shows a relatively weak viscosity-enhancing effect. Therefore, GPP/SEBS exhibits high zero shear viscosity and low shear viscosity. High viscosity at low shear frequency is beneficial to control the eccentricity of large cross-section cable insulation, whereas low shear viscosity is beneficial to cable extrusion.

In the case of DC electrical properties, the presence of polystyrene units in the molecular structure of SEBS can introduce charge traps slightly deeper than the intrinsic trap energy level at the interface, and capture the high-energy electrons injected by the electrode under the action of an electric field. Thus, the probability of electron injection into the intrinsic trap is reduced. However, this ability is limited because the SEBS phase cannot be uniformly dispersed in the matrix, so the space charge suppression effect of PP/SEBS material is not ideal. The grafted polar groups can introduce deeper charge traps at the high temperature side, and the intrinsic charge release peak of the GPP/SEBS sample at 80°C is obviously inhibited. Meanwhile, there are two new charge release peaks near 101°C and 130°C, which are the deep traps introduced by polystyrene units and grafted polar functional groups in SEBS. In the GPP/SEBS composites, the size of the dispersed phase of SEBS is significantly reduced, which reduces the large-size free volume in the material and reduces the generation of macroscopic defects. In addition, steric hindrance reduces the mobility of macromolecular chain segments and side groups and inhibits the migration of carriers. The traps introduced by polystyrene structural units at the two-phase interface have significant trapping and scattering effects on carriers, which play a trapping mechanism together with the deep traps introduced by grafting modification. It can be used as a hole or an electron vacancy to capture the moving charge inside the material and effectively reduce the mobility of the carrier. Meanwhile, the deep trap mechanism introduced by the grafted MAH polar group can effectively reduce the conductance current of the composite under electric field, and the DC dielectric properties of the composites are improved obviously [22–26].

5 | Conclusions

In view of PP/SEBS HVDC cable insulation materials, GPP was prepared by grafting functional polar groups onto PP using UV light grafting technology. The mechanical properties and viscoelastic properties of the composites were enhanced by improving the phase dispersion of elastomer SEBS in

GPP/SEBS. Additionally, the introduction of polar groups created charge traps that elevated the DC electrical performance. The properties of PP, GPP, PP/SEBS and GPP/SEBS were characterised and compared. The main conclusions are as follows:

1. SEM results and compatibility calculation results show that the compatibility between GPP and SEBS is greatly improved compared with PP after grafting modification, and the size of elastomer phase in GPP/SEBS is reduced. Mechanical and thermal properties results show that compared with PP/SEBS, GPP/SEBS with smaller SEBS size shows better mechanical properties and better soften effect. In addition, GPP/SEBS exhibits larger zero shear viscosity and smaller shear viscosity, which is beneficial to cable processing.
2. DC electrical properties results show that compared with PP, the space charge characteristics of PP/SEBS are improved, the conductivity characteristics are not obviously improved and the DC breakdown strength is significantly reduced. After grafting modification, GPP/SEBS shows excellent space charge, suppressed conductance current, and enhanced DC breakdown strength.
3. TSDC and DOS results show that SEBS can introduce shallow charge traps in PP, which exhibits a limiting effect on charge transport at low electric field. After graft modification, deep charge traps appear in GPP. These deep traps can effectively regulate carrier transport and improve electrical properties. GPP/SEBS has the charge traps introduced by SEBS and GPP, which is consistent with its excellent DC properties.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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